

MATH – 241
Engineering Mathematics I
Problem Set – 3

Question 1. Let $W = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\}$. Show that W is subspace of $M_2 \subset \mathbb{R}$ over $\mathbb{R}$ and that the following matrices form a basis for W . (M_2 is the set of all 2×2 matrices with real entries.)

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Find the coordinates which are the representation coefficients of the matrix $\begin{bmatrix} 1 & -2 \\ -2 & 3 \end{bmatrix}$ under the basis.

Question 2. Let V be a finite dimensional vector space and S be a subspace of V . Show that

- (a) $\dim S \leq \dim V$.
- (b) $\dim S = \dim V$ if and only if $S = V$.
- (c) Every basis for S is contained in some basis for V .
- (d) A basis for V need not contain a basis for S .

Question 3. Let U and V be subspaces of an n -dimensional vector space S having dimensions p and q respectively. Let $W = \{\underline{w} : \underline{w} = \underline{u} + \underline{v} \text{ for a } \underline{u} \in U \text{ and } \underline{v} \in V\}$:

- (a) Show that W is a subspace of S .
- (b) Is the representation of a vector in W as $\underline{w} = \underline{u} + \underline{v}$ always unique? If not, find the condition that it is unique.
- (c) Show that $\dim W \leq \min\{n, p + q\}$.

Question 4. Prove or give a counterexample : If U_1, U_2, W are subspaces of V such that $U_1 + W = U_2 + W$ then $U_1 = U_2$.

Question 5. Let W_1 and W_2 be nontrivial subspaces of a vector space V ; that is, neither $\{0\}$ nor V . Show that there exists an element $\alpha \in V$ such that $\alpha \notin W_1$ and $\alpha \notin W_2$. Show further that there exists a basis of V such that none of the vectors in the basis is contained in either W_1 or W_2 . Is this true for more than two nontrivial subspaces?

Question 6. Let V be a vector space. Prove that, if $\{v_1, v_2, \dots, v_n\}$ is a basis of V then so does

$$\{v_1 - v_2, v_2 - v_3, \dots, v_{n-1} - v_n, v_n\}$$